



Organ-level instance segmentation enables continuous time-space-spectrum analysis of pre-clinical abdominal photoacoustic tomography images

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ABSTRACT

Photoacoustic tomography (PAT), as a novel biomedical imaging technique, is able to capture temporal, spatial and spectral tomographic information from organisms. Organ-level multi-parametric analysis of continuous PAT images are of interest since it enables the quantification of organ specific morphological and functional parameters in small animals. Accurate organ delineation is imperative for organ-level image analysis, yet the low contrast and blurred organ boundaries in PAT images pose challenge for their precise segmentation. Fortunately, shared structural information among continuous images in the time-space-spectrum domain may be used to enhance segmentation. In this paper, we introduce a structure fusion enhanced graph convolutional network (SFE-GCN), which aims at automatically segmenting major organs including the body, liver, kidneys, spleen, vessel and spine of abdominal PAT image of mice. SFE-GCN enhances the structural feature of organs by fusing information in continuous image sequence captured at time, space and spectrum domains. As validated on large-scale datasets across different imaging scenarios, our method not only preserves fine structural details but also ensures anatomically aligned organ contours. Most importantly, this study explores the application of SFE-GCN in multi-dimensional organ image analysis, including organ-based dynamic morphological analysis, organ-wise light fluence correction and segmentation-enhanced spectral un-mixing. Code will be released at <https://github.com/lzc-smu/SFEGCN.git>.

1. Introduction

Photoacoustic tomography (PAT) is an emerging imaging modality that has been used in preclinical research (Ivankovic et al., 2019; Knieling et al., 2017; Li et al., 2019; Taruttis and Ntziachristos, 2015; Tang et al., 2023) owing to its superior spatial resolution, rich imaging contrast, and non-ionizing properties of light radiation. As a multispectral imaging technique, PAT enables the acquisition of spatial, temporal, and spectral information from biological tissues (Li et al., 2022; Zhang et al., 2023; Razansky et al., 2011), encompassing applications in cardio-vascular and inflammatory diseases (Deán-Ben et al., 2015), blood oxygen saturation (sO₂) visualization (Li et al., 2018), tumor localization (Zhang et al., 2019a), organ damage analysis (Lv et al., 2021) and drug delivery (Zhang et al., 2019b; Manivasagan et al., 2016). The multi-dimensional continuous imaging capability of PAT is illustrated in Fig. 1 as three representative scenarios: longitudinal observation (Fig. 1b), multi-slice axial scanning (Fig. 1c), and multispectral imaging (Fig. 1d). In longitudinal observations, i.e., time-domain imaging,

the image sequence reflects continuous changes in tissue movement, morphology, and function of organs over time. In axial scanning scenarios, i.e., space-domain imaging, a volumetric model of the imaged target can be obtained. In multispectral imaging, i.e., spectrum-domain imaging, PAT images excited by light at multiple wavelengths is used to distinguish the distribution of optical absorbers within tissue, such as oxygenated/deoxygenated hemoglobin (HbO₂ and Hb).

In abdominal PAT imaging of small animals, the above multi-dimensional imaging applications often require organ level image analysis. It involves the extraction of quantitative information such as organ shape and size, chromophore concentration and localization, drug deposition, signal distribution of endogenous and exogenous biomarkers, etc. These multi-parametric image analysis tasks make accurate delineation of organ anatomy necessary. For example, in longitudinal imaging, organ morphological changes along time can be quantified with the help of organ segmentation (Fig. 1f). Moreover, segmentation of different organs may further enable organ-level image processing.

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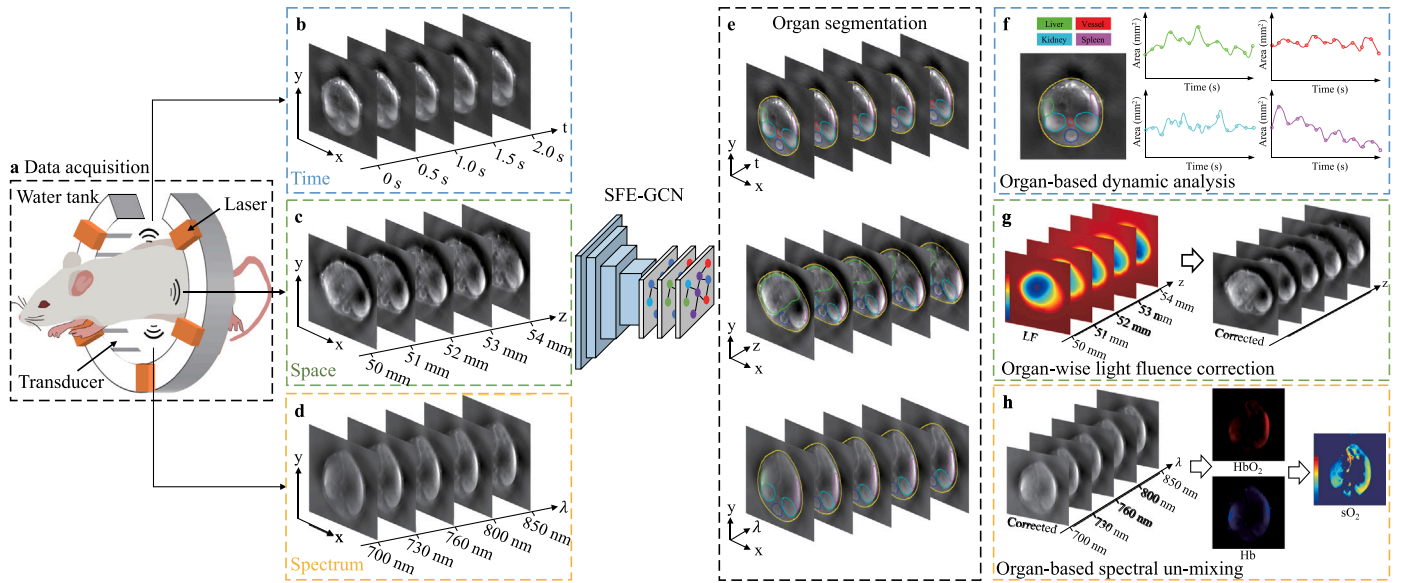


Fig. 1. Schematic diagram of abdominal organ segmentation and analysis of pre-clinical PAT images. a: Data acquisition: continuous PAT images of mouse abdominal organs are acquired. b: Time-domain longitudinal observation. c: Space-domain multi-slice axial scanning. d: Spectral-domain multispectral imaging. e: Multi-organ instance segmentation results. f: Organ-based dynamic morphological analysis. g: Organ-wise light fluence correction. Corrected: light fluence corrected PAT image. h: Organ-based spectral un-mixing. SFE-GCN: structure fusion enhanced graph convolutional network. HbO₂: oxygenated hemoglobin, Hb: deoxygenated hemoglobin, sO₂: blood oxygen saturation.

Since the optical energy density distribution provided by the PAT image is the product of the absorption coefficient and the light fluence (LF) distribution (Cox et al., 2012), correcting for LF attenuation is an effective means to enhance image quality and recover the distribution of optical absorbers (Cox et al., 2006; Zhang et al., 2022b). Current LF correction methods assume that the entire animal body has uniform optical absorption, and thus only segment the body surface (Liang et al., 2024, 2022; Mandal et al., 2016; Zhang et al., 2022a). Their correction performance is limited because they ignore the optical heterogeneity of different organs. Therefore, given the accurate segmentation of major organs, the optical absorption parameters of these organs can be introduced into LF correction to enhance performance (Fig. 1g). Furthermore, combined with LF correction at organ-level, un-mixing of multi-wavelength PAT images may improve the localization of light absorbers in deep tissue (Fig. 1h).

Currently, organ segmentation for PAT image is performed manually by identifying the organ contour in each slice, which is highly repetitive and time-consuming. The challenge escalates in continuous imaging where a larger number of images is acquired. Furthermore, the accuracy and reproducibility of manual segmentation are highly dependent on the operator's experience and have large inter-observer variability (Liang et al., 2022). To overcome these problems, several automatic image segmentation methods in PAT have been reported, e.g. automatic recognition of structures in small animal images (Liang et al., 2024, 2022; Mandal et al., 2016), tumor segmentation (Zhang et al., 2019a), blood vessel segmentation (Yuan et al., 2020) and skin layer segmentation (Ly et al., 2022; Moustakidis et al., 2019). Previously, our group has also proposed a surface extraction method for small animal body (Liang et al., 2024). It is based on graph convolution network (GCN) and is employed for 3D continuous image segmentation. However, these works usually focus on one specific organ or tissue type in local imaging region. Schellenberg et al. (2022) introduced a multi-objective semantic segmentation method for hybrid optoacoustic/ultrasound images. Based on linear transducer PAT images, the method effectively distinguishes skin, blood, and fat on the superficial human skin tissue. Except for this, there is no method for the instance segmentation of a large number of organs for PAT images so far.

The typical characteristic of small animal abdominal organs in PAT images is marked by low imaging contrast, blurred boundaries, and morphological complexity, thus posing challenges to accurate organ

segmentation. Furthermore, the substantial variations in the relative sizes of different organs e.g., the liver compared to the vessel, contribute to the vulnerability of traditional segmentation methods based on convolutional neural network (CNN), resulting in the potential loss of local semantic information. Compared to methods based on CNN (Hao and Li, 2023; Marin et al., 2019), GCN-based approaches demonstrate efficacy in mitigating the loss of position and structure information during feature extraction, yielding smooth contours consistent with authentic anatomical features (Liang et al., 2024). This methodology has found widespread application in medical image segmentation tasks, encompassing fuzzy nodule boundary segmentation in ultrasound images (Huang et al., 2022), prostate segmentation in MRI images (Tian et al., 2020), and polyp segmentation in colonoscopy images (Meng et al., 2021). Consequently, the incorporation of this method into the task of organ segmentation in PAT images is deemed feasible. Moreover, in multi-dimensional continuous PAT imaging, the acquired image sequence shares abundant information of the imaged subject. The spatial-temporal-spectral information may be induced to the model for enhancing the precision of image segmentation.

Based on the above analysis, we present a multi-organ instance segmentation pipeline based on an end-to-end structure fusion enhanced GCN for pre-clinical abdominal PAT image. Our new method, which we named SFE-GCN, incorporates a contextual fusion of structural feature to extract the spatial-temporal-spectral associative information provided by continuous images. We used a CNN-based backbone to extract semantic features and then generate initial contours and classification attributes for each organ. To mitigate the loss of positional and structural information, we employ a residual GCN (ResGCN) to estimate the vertex-wise offset of the contour to obtain the final segmentation boundary. Without any human intervention or parameter-tuning, our method is able to automatically segment major organs including the body, liver, kidneys, spleen, vessels and spine in a set of abdominal PAT images, whether it is spatially, temporally, or spectrally continuous. We validate its performance on over 3000 annotated mouse abdominal PAT images, representing the largest reported validation of mouse abdominal organ segmentation. More importantly, we affirm the advantages of our segmentation method in multi-dimensional organ-level analysis tasks pertaining to organ dynamic analysis, organ-based LF attenuation correction, and enhanced visualization of endogenous contrast agents in deep tissues.

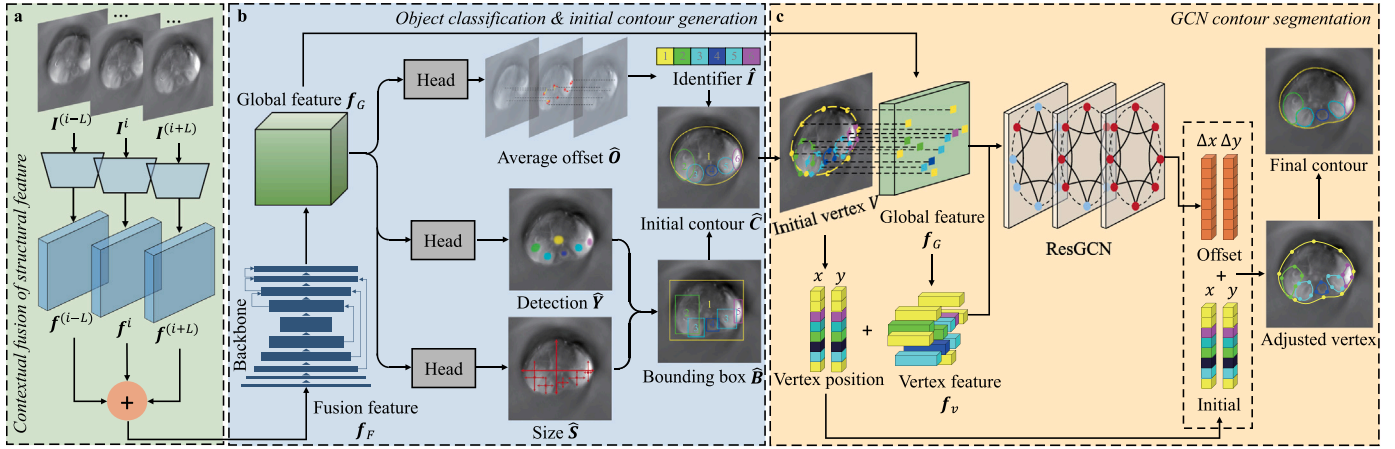


Fig. 2. Overview of the proposed SFE-GCN. a: Contextual fusion of structural feature: low-level semantic information is fused in continuous images. b: Object classification and initial contour generation: multi-scale feature information is extracted and classification bounding boxes are obtained to generate initial contours. c: GCN contour segmentation: the ResGCN module utilizes global structural information of vertex boundaries to predict the final contour.

In summary, this study makes the following contributions.

- (1) SFE-GCN is the first solution for accurate multi-organ instance segmentation in abdominal PAT images.
- (2) Our method utilizes the shared structural information of continuous PAT images to improve segmentation accuracy.
- (3) We evaluate the performance of our SFE-GCN method on time, space, and spectrum domain continuous abdominal PAT datasets with more than 3000 images.
- (4) Enabled by multi-organ segmentation, we demonstrate quantitative analysis of time-varying organ size changes, organ-wise LF correction, and enhanced visualization of spectrally unmixed light absorbers.

The rest of the paper is organized as follows. In Section 2, we present the overall framework of the SFE-GCN and provide detailed descriptions of its components. Section 3 elaborates on the experimental data, parameter settings, and quantitative evaluation metrics. Section 4 provides a comprehensive analysis of the experimental results. Finally, discussion and conclusion are given in Section 5 and Section 6, respectively.

2. Method

2.1. Overall framework

The architecture of the proposed SFE-GCN framework is shown in Fig. 2. Given a continuous image sequence (from the time, space or spectrum domain) as input, the method first extracts low level features from the image sequence and fuses them through summation. Then, the multi-scale CNN feature extractor extracts the semantic feature at different scales to produce a global feature map. Three prediction heads are employed to assign classification labels and generate initial contours for each organ. Finally, the ResGCN module transforms the initial contours into graph-structured data and then predicts the offsets for all vertices to generate the final segmentation contours.

2.2. Contextual fusion of structural features

In PAT images acquired through continuous scanning, strong structural similarity exists among the spatial, temporal, and spectral imaging domains. Leveraging this similarity, we can enhance discriminative information for target organs and thus improve abdominal organ segmentation accuracy. In SFE-GCN, we employ a straightforward summation fusion strategy. For a given target image sequence $S = \{I^{i-L}, \dots, I^i, \dots, I^{i+L}\}$ with a sampling step L (using blank frame padding to complete the sequence), the low-level structure feature of

each image $I \in \mathbb{R}^{H \times W}$ is encoded by a weight-shared coding unit comprising a 2D Conv (kernel Size: 7; padding: 3; stride: 1), batch normalization and rectified linear activation function. Here, H and W represent the height and width of the image. The resulting feature matrices $F = \{f^{i-L}, \dots, f^i, \dots, f^{i+L}\}$ are then summed to generate the fusion feature map f_F (Fig. 2a), where $f^i \in \mathbb{R}^{H \times W \times 16}$.

2.3. Classification and contour initialization

The generation of the initial contour can be reframed as an object detection task, involving category determination, positioning, and size estimation. Inspired by Zhou et al. (2019), we use a deep layer aggregation (DLA) as the network backbone (Fig. 2b). Taking the fusion feature f_F as input, the DLA backbone extracts and fuses global structural feature and deep semantic feature, yielding a global feature map f_G . Subsequently, a prediction head generates the detection central heatmap $\hat{Y} \in \mathbb{R}^{\frac{H}{R} \times \frac{W}{R} \times C}$ from the global feature, where C represents the number of organ types and R is the output stride. Each peak $\hat{p} = [x_{\hat{p}}, y_{\hat{p}}]$ in the heatmap $\hat{Y}$ corresponds to a detected object center with confidence $\hat{Y}_{\hat{p}}$ (different organ categories are represented by different colors in Fig. 2b). For each ground truth detection point p , a low-resolution equivalent can be computed by $\tilde{p} = \frac{p}{R}$. The ground truth heatmap Y is generated by using the Gaussian kernel $Y = \exp(-\frac{(x-\tilde{p}_x)^2 + (y-\tilde{p}_y)^2}{2\sigma_p^2})$, where σ_p is the adaptive standard deviation. Finally, the focus loss is applied for training (Lin et al., 2020):

$$L_{detect} = -\frac{1}{N} \sum_{xyz} \begin{cases} (1 - \hat{Y})^\alpha \log(\hat{Y}), & \text{if } Y = 1 \\ (1 - \hat{Y})^\beta (\hat{Y})^\alpha \log(1 - \hat{Y}), & \text{otherwise} \end{cases}, \quad (1)$$

where N is the number of organs, and α and β are hyper-parameters of the focal loss. We set $\alpha = 2$ and $\beta = 4$ following Law and Deng (2020).

Another regression head outputs a size map $\hat{S} \in \mathbb{R}^{\frac{H}{R} \times \frac{W}{R} \times 2}$ of bounding boxes for all objects. The size prediction is only supervised at the central location $\hat{p}_k$:

$$L_{size} = \frac{1}{N} \sum_{k=1}^N |\hat{S}_{\hat{p}_k} - S_{\hat{p}_k}|, \quad (2)$$

where $S_{\hat{p}_k}$ is the ground truth size of the k th organ at position $\hat{p}_k$. By combining the location and category information from the detection central heatmap $\hat{Y}$ with the size information from the size heatmap $\hat{S}$, we can determine the bounding boxes $\hat{B}$ for each organ. The bounding box of k th organ at the peak $\hat{p}_k$ is generated using the detection central $\hat{Y}_k = [x_{\hat{p}_k}, y_{\hat{p}_k}]$ and the size $\hat{S}_k = [w_{\hat{p}_k}, h_{\hat{p}_k}]$:

$$\hat{b} = [x_{\hat{p}_k} - \frac{w_{\hat{p}_k}}{2}, y_{\hat{p}_k} - \frac{h_{\hat{p}_k}}{2}, x_{\hat{p}_k} + \frac{w_{\hat{p}_k}}{2}, y_{\hat{p}_k} + \frac{h_{\hat{p}_k}}{2}]. \quad (3)$$

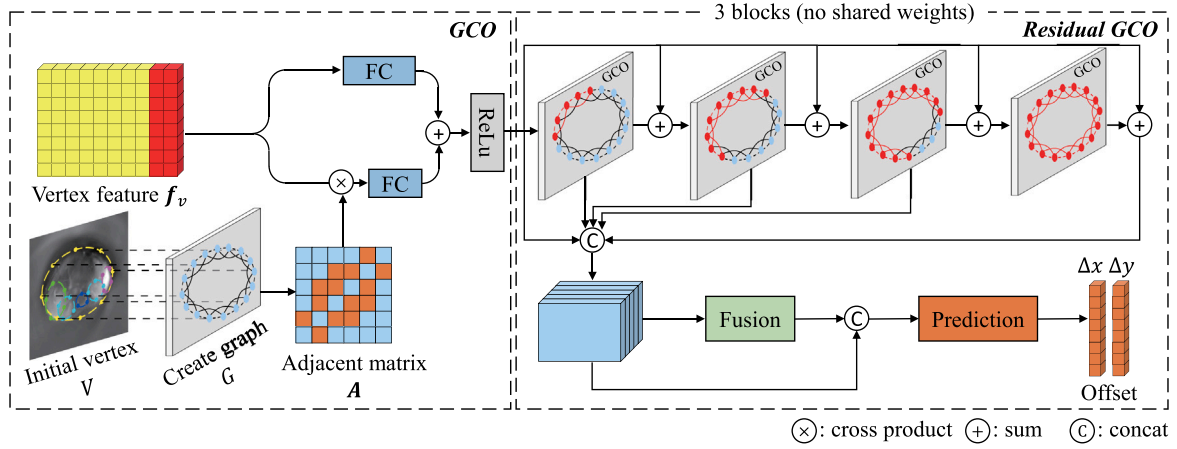


Fig. 3. Schematic diagram of Residual GCN. The GCN consists of multiple residual graph convolution operations (GCO). Vertex feature: the part from the global feature map is marked in yellow, and the current position of the vertex is marked in red. FC is a fully connected layer.

Finally, the standard elliptical initial contours $\hat{C}$ are generated based on the predicted bounding boxes $\hat{B}$. The ellipse is characterized by:

$$\frac{(x - x_{\hat{p}_k})^2}{(w_{\hat{p}_k}/2)^2} + \frac{(y - y_{\hat{p}_k})^2}{(h_{\hat{p}_k}/2)^2} = 1. \quad (4)$$

In the context of data analysis tasks such as organ motion and morphological change, the assignment of a unique label to each organ becomes imperative. In typical PAT imaging experiment, the minimal position variations of the same organ between adjacent slices serve as valuable reference information for assigning identifiers. To establish correlation between positions of consecutive detection points, we adopt the methodology presented in Zhou et al. (2020) by incorporating an additional prediction head. This head supplies the average offset $\hat{O}^i \in \mathbb{R}^{\frac{H}{K} \times \frac{W}{K} \times 2}$ for the same organ at detection central points. For each organ detected at position $\hat{p}^i$, the offset $\hat{o}^i = \hat{O}^i_{\hat{p}^i}$ signifies the average offset of the organ in the imaging sequence: $\hat{o}^i = \frac{1}{2L} \sum_{j=i-L}^{i+L} (\hat{p}^j - \hat{p}^i)$. This average offset is learned using the same regression objective as for size prediction:

$$L_{off} = \frac{1}{N} \sum_{k=1}^K |\hat{O}^i_{\hat{p}^i_k} - \frac{1}{2L} \sum_{j=i-L}^{i+L} (p^j_k - p^i_k)|, \quad (5)$$

where p_k is the central position of ground truth organ. Based on average offset, organs can be associated using a greedy matching algorithm (Zhou et al., 2020). For each detection at position $\hat{p}$, it will be greedily associated with the closest unmatched prior detection at position $\hat{p} - \hat{o}$. If there is no unmatched prior detection within radius κ , a new identifier is assigned. In this way, the identifier $\hat{I}$ assigns a unique identifier to each organ, distinguishing every organ instance within the initial contour $\hat{C}$ (Fig. 2b).

2.4. GCN-based contour delineation

Recently, the application of the ResGCN to body segmentation in PAT images has been explored (Liang et al., 2024). Experimental finding demonstrates that ResGCN effectively captures the topological correlation within imaging tissues, resulting in continuous and smooth segmentation contours. Motivated by this, we adopt the framework based on ResGCN for abdominal multi-organ segmentation. In Fig. 2c, we select K points from the elliptical initial contour to create a list of contour vertices $V = v_0, v_1, \dots, v_{K-1}$. Each vertex's single-channel vector is computed using bilinear interpolation at the corresponding position on the global feature map f_G and connected to its current position (x, y) , serving as the vertex feature input f_v for ResGCN. To expedite information exchange among vertices, we establish a closed

cylindrical graph topology by connecting each vertex to its 1- and 2-neighborhood vertices, as illustrated in Fig. 3. Formally, we define the graph as $G = (V, E)$, where V represents the vertices and E denotes the edges connecting two adjacent vertices in the graph, and its connection structure is represented by the adjacency matrix A .

Fig. 3 provides a detailed schematic diagram of the ResGCN. Both the vertex feature matrix f_v and the adjacent matrix A serve as inputs to the graph convolution operation (GCO). In each GCO, vertex feature are updated by aggregating the features of neighboring vertices. Mathematically, if $f_{gc}^j \in \mathbb{R}^{M \times d_j}$ denotes the j th layer feature updated by the graph convolution, where M is the number of the vertices on the initial contour and d_j is the feature dimension of each vertex, the information propagation formula of GCO is expressed as (Liang et al., 2024):

$$f_{gc}^{j+1} = \text{ReLU}(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} w^j f_{gc}^j), \quad (6)$$

where $\tilde{A} = A + I_N$ is the adjacency matrix with self-connections achieved by adding the identity matrix I_N . $\tilde{D}$ is the degree matrix of $\tilde{A}$, and w^j is a weight matrix.

We form a Residual GCO block consisting of five cascaded GCOs, using residual skip connections for all layers to propagate information among vertices within the graph. The output of each GCO is interconnected via skip connection and fused using a fusion block containing 1×1 Conv and max pooling. The resulting feature is then concatenated with the output of each layer to predict the vertex-wise offsets $(\Delta x, \Delta y)$ through a dedicated prediction head. In ResGCN, three cascaded Residual GCO blocks generate the final vertex offsets. This module is trained by a key vertex loss (Peng et al., 2020). Both the predicted and ground truth vertices are sampled to K points in counterclockwise order, with $K=64$ representing the contour positions. Assuming equally sized and similarly ordered prediction and ground truth vertex sets, denoted as $vp = \{vp_0, vp_1, \dots, vp_{K-1}\}$ and $vg = \{vg'_0, vg'_1, \dots, vg'_{K-1}\}$, respectively. The loss function is calculated as follows:

$$L_{vertex}(vp, vg) = \frac{1}{K} \sum_{i=0}^{K-1} \|vp_i - vg'_i\|_1. \quad (7)$$

Overall, in each iteration, the training objective is defined as:

$$L = L_{detect} + \lambda_s L_{size} + \lambda_o L_{off} + \lambda_v L_{vertex}, \quad (8)$$

where $\lambda_s=0.1$, $\lambda_o=1$ and $\lambda_v=1$ are scaling factors.

3. Experiments

3.1. Datasets

We utilized three abdominal PAT datasets of nude mouse, respectively obtained through longitudinal temporal observation (time domain), multi-slice axial scanning (space domain), and multispectral

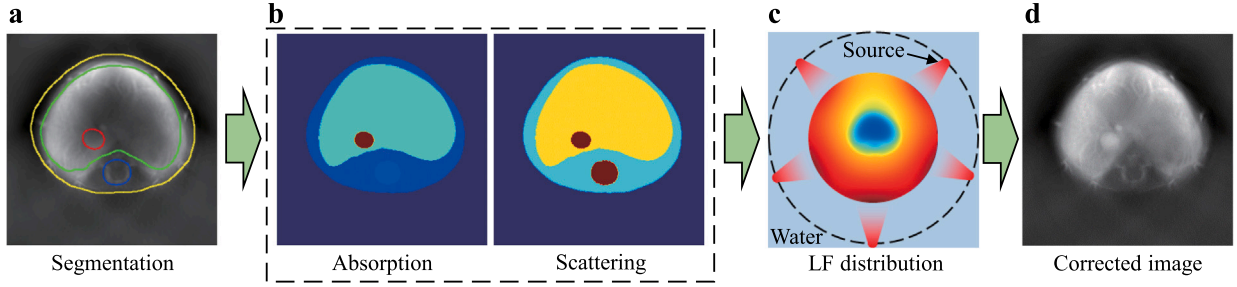


Fig. 4. Light fluence estimation and correction procedures. a: Multi-organ segmentation. b: Absorption and scattering maps for each region. c: LF distribution map. d: LF corrected PAT image.

Table 1

Details of the time domain (TD), spatial domain (SD), and spectral domain (WD, W for wavelength) datasets used in our experiments.

Dataset	Time interval (s)	Space interval (mm)	Wavelength (nm)	Split (train/val/test)
TD dataset	0.5	/	760	310/30/60
SD dataset	/	1	760	990/60/150
WD dataset	/	1	700/730/760/800/850	1200/100/375

imaging (spectrum domain), to train and evaluate the proposed SFE-GCN model. These datasets are partitioned into training, validation, and testing sets. Table 1 summarizes the image parameters of the time domain (TD), space domain (SD) and spectrum domain (WD, W for wavelength) datasets. Image acquisition is conducted using a commercial small animal multispectral PAT system (MSOT inVision128, iThera Medical, Germany), as extensively described in Liang et al. (2022), Qi et al. (2021). The laser pulses of our MSOT device have a duration of approximately 5 ns and a repetition rate of 10 Hz, allowing the shortest time interval to be controlled at 0.1 s. The spacing between spatial measurements is precisely controlled by the motorized linear translation stage. The laser wavelength is tunable from 680 nm to 980 nm.

All experimental procedures adhere to institutional guidelines and receive approval from Southern Medical University. Transverse images are standardized to a 512×512 pixel scale for training, and the reference segmentation contours are meticulously annotated by two trained analysts. Under the same segmentation guidelines, the analysts are given the opportunity to learn and perform organ segmentation on example PAT images before starting the annotation. Spline interpolation was then performed based on these landmarks to obtain initial contours. The analysts can continue to adjust the landmarks to refine the contours until satisfactory. The regions where the two analysts reached consensus after collaborative correction are regarded as segmentation reference.

Due to differences in acquisition intervals and ranges, we adopt different grouping methods for the three domain datasets as inputs for SFE-GCN. For the TD dataset, PAT images collected within 2.5 s are grouped as a set. For the SD dataset, axial scanned PAT images within a 15 mm range are grouped as a set. For the WD dataset, PAT images of the same cross-section collected at five specific wavelengths are grouped as a set. For images within the same group, we augment input images and annotations with the same random translation, rotation, and scaling. To enable the model to quickly learn basic features of PAT images, we first pre-trained the SFE-GCN using all training data (train: 2500, val: 190) from the three domains and then fine-tuned the pre-trained model using the TD, SD, and WD datasets separately to obtain three SFE-GCN models tailored for different imaging scenarios.

3.2. FEM-based light fluence estimation

The reconstructed PAT image at each wavelength provides a spatial map of absorbed light energy within imaging tissue (Cox et al., 2012). Assuming the thermo-elastic Grüneisen parameters is constant, a PAT image $p' = \Phi(\mu_a, \mu_s')$ can be expressed as the product of the

LF distribution Φ and the absorption μ_a and the reduced scattering coefficient μ_s' (Zhang et al., 2022b). LF correction involves computing the LF distribution of the laser light across the entire imaging object and subsequently dividing the PAT image by this LF map, as shown in Fig. 4. Assuming uniform optical coefficients within a single organ, the absorption and scattering maps (Fig. 4b) for each region can be explicitly defined after obtaining multi-organ segmentation (Fig. 4a) using SFE-GCN. Given the distribution of optical absorption and scattering of different organs, the light fluence can be simulated according to the photon Diffusion Equation (DE) (Liang et al., 2022). This is done by using Finite Element Method (Fig. 4c), with the open-source Toast++ toolbox (Schweiger and Arridge, 2014). During optical transport modeling, the background medium is set to water, and the absorption and scattering coefficients of all organs were approximated based on optical coefficients of biological tissue measured in Alexandrakis et al. (2005), as listed in Table 2. The “Body” region in this paper is approximated using muscle coefficients, while the coefficients for the “Vessel” class are defined using the whole blood parameters with an sO_2 level set to 75% (Alexandrakis et al., 2005).

3.3. Spectral un-mixing

Spectral un-mixing involves identifying variations in the spectrum domain. Each pixel in the PAT image encompasses a mixture of various absorbers, discernible through individual images at each wavelength. In this study, we employ the linear un-mixing method (Keshava and Mustard, 2002) to calculate the distribution of HbO_2 and Hb from the normalized PAT images. Once the concentrations of HbO_2 (c_{HbO_2}) and Hb (c_{Hb}) are determined, the oxygen saturation level sO_2 can be computed as (Zhang et al., 2022a):

$$sO_2 = \frac{c_{HbO_2}}{c_{HbO_2} + c_{Hb}}. \quad (9)$$

3.4. Comparative experiments

To validate the efficacy of the proposed SFE-GCN framework, we conducted comprehensive comparisons with established methods, including the powerful baseline nnU-Net model (Isensee et al., 2021), the mask-based instance segmentation method PointRend (Kirillov et al., 2020) and the GCN-based instance segmentation model PolySnake (Feng et al., 2023). Furthermore, to assess the performance of the proposed model in the context of LF correction experiment for multi-organ segmentation, we use a reference method that only segments body region, assuming uniform optical coefficients for the entire body.

Table 2

Approximated absorption and scattering coefficients for each organ used in this work (not real measured values).

Wavelength (nm)	Compiled μ_a (mm ⁻¹)					Compiled μ'_s (mm ⁻¹)				
	700	730	760	800	850	700	730	760	800	850
Body	0.014	0.009	0.012	0.008	0.008	0.38	0.34	0.30	0.26	0.22
Liver	0.115	0.078	0.105	0.063	0.063	0.65	0.62	0.59	0.56	0.52
Kidney	0.021	0.015	0.020	0.012	0.012	2.11	1.98	1.86	1.72	1.57
Spleen	0.115	0.078	0.105	0.063	0.063	0.65	0.62	0.59	0.59	0.52
Vessel	0.385	0.259	0.349	0.211	0.209	1.76	1.71	1.67	1.61	1.55
Spine	0.020	0.013	0.018	0.010	0.010	2.34	2.20	2.07	1.92	1.76
Water	0.00001					0.00001				

We performed LF correction experiments using SD and WD datasets, and compared the PAT image quality before and after correction qualitatively and quantitatively. Finally, we performed spectral un-mixing experiments using the multispectral PAT images in WD dataset and compared the spectral un-mixing results of our organ-wise correction method and body-only correction method. Since the segmentation performance of SFE-GCN is independent across the sequential PAT images, i.e., the segmentation result on the current slice does not affect the performance on adjacent sequential PAT images. Thus, we treated each organ region in the PAT image as an independent segmentation sample for both qualitative and quantitative analyses.

3.5. Evaluation metrics

For quantitative segmentation performance evaluation, we employ the Matthews Correlation Coefficient (MCC) as well as the 95% Hausdorff Distance (HD₉₅) (Liang et al., 2024). In addition, to quantitatively evaluate the impact of LF correction experiments, we measure the Coefficient of Variation (CV) and Contrast to Noise Ratio (CNR) of the image before and after correction (Liang et al., 2022). The results of quantitative comparative analysis in this paper are expressed as mean $\pm$ 95% confidence intervals.

For the MCC score, a pixel is classified as a true positive (TP) if correctly predicted as part of the target organ, as a false positive (FP) if incorrectly predicted as part of the target organ, as a true negative (TN) if correctly predicted as background, or false negative (FN) if incorrectly predicted as background. The MCC is then calculated as:

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} + \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}, \quad (10)$$

and HD₉₅ quantifies the distances between the surfaces of the predicted segmentation and the reference annotation for a given organ. It is expressed as:

$$\text{HD}_{95} = \max_{\hat{c} \in \hat{C}} \left(\max_{c \in C} \|\hat{c} - c\| \right), \max_{c \in C} \left(\max_{\hat{c} \in \hat{C}} \|\hat{c} - c\| \right), \quad (11)$$

where $\hat{C}$ is the predicted contour, and C is the human reference annotation. For methods that do not explicitly output contours, we used the *mask2poly* tool (Nikolay, 2024) to extract the outer contour of the segmentation mask.

CV is an index to measure the signal intensity non-uniformity, while CNR reflects an objective measure of useful contrast between background and target. They are given by:

$$\text{CV} = \frac{\sigma_t}{\mu_t}, \quad (12)$$

$$\text{CNR} = 10 \log \left(\frac{\mu_t - \mu_b}{\sqrt{\sigma_t^2 + \sigma_b^2}} \right), \quad (13)$$

where σ_t and σ_b denote the standard deviation of the target and background areas of the PAT image. μ_t and μ_b are the pixel averages of the target and background region, respectively.

3.6. Implementation details

The SFE-GCN is implemented in Python with developed codes subject to the PyTorch framework. The algorithm runs on an Ubuntu system with an Intel Xeon E5-2667 CPU (3.2 GHz) with 64 GB memory. Our network is trained end-to-end for 200 epochs on an NVIDIA TITAN X Pascal GPU with 12 GB memory. Model training for one epoch takes between 5 to 6 min (depending on the size of dataset). Adam optimizer is used to update the weights. The weight decay is set to 5e-4 and the momentum constant is set to 0.9. During training, the learning rate starts from 1e-4 and decreases by 0.5 at 50, 100 and 150 epochs.

4. Results

4.1. Results on time domain continuous PAT images

Segmentation performance. The qualitative results of four different segmentation methods on TD dataset are shown in Fig. 5a. It can be seen that our method demonstrates results closest to the reference contours among the compared models. In the other methods, the small vessel cannot be correctly detected and segmented (white arrows), whereas our method is able to accurately obtain contours consistent with human reference annotation. It demonstrates that SFE-GCN can avoid the loss of the information of small structure. The result show that some parts of the body masks produced by nnU-Net and PointNet extend into the background, which might indicate their susceptibility to background noise. In addition, while the PolySnake model performs well in segmentation, it is susceptible to classification errors (e.g., the spine being misclassified as a vessel, and the right kidney being misclassified as the spleen). We quantitatively compare the segmentation accuracy of SFE-GCN with the other methods for each organ on the TD dataset, and the result is listed in Table 3. From the quantitative results, we can observe that our method outperforms all other comparison methods in terms of both MCC and HD₉₅ scores across all organs. For the kidneys with relatively weak boundaries, the performance of PolySnake and SFE-GCN is significantly improved, indicating the advantage of the graph convolutional network in handling boundary structure. In terms of overall average score, SFE-GCN yields the highest MCC of 93.44% (95% confidence interval: [93.20, 93.68]) and the lowest HD₉₅ of 0.51 mm (95% confidence interval: [0.50, 0.53]). In summary, these results underscore that our method either matches or exceeds the state-of-the-art performance on TD dataset.

Organ-based dynamic analysis. Since our method assigns a unique label to each individual organ, organ-based dynamic analysis tasks such as motion tracking, sO₂ monitoring, and organ morphological changes can be easily implemented in the continuous-time observed scanning images. Utilizing the SFE-GCN segmentation results, we conduct multi-organ motion analysis on PAT images observed at five time points within the TD dataset (Fig. 5b). The current position of the organs is displayed with red dots, while the direction and trend of organ motion is indicated by red arrows with the size of arrow proportional to the extent of organ movement. It can be found that larger organs (kidneys and spine) exhibit minimal shifts, whereas the small vessel and spleen near the body surface display frequent movements due to

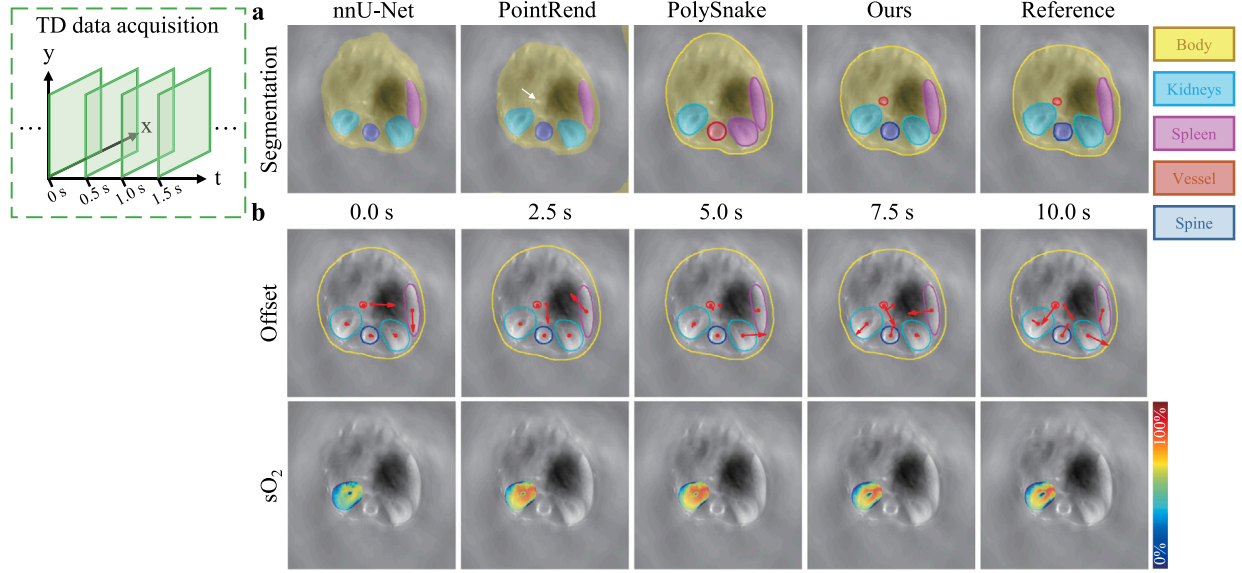


Fig. 5. Results of time-continuous imaging experiments . a: Qualitative comparison of four segmentation methods on the TD dataset. b: Top: organ-based motion offset analysis. The red arrow points to the offset direction, and its size represents the offset trend. Bottom: Changes in sO_2 distribution of the left kidney at different time. Scale bar: 5 mm.

Table 3

Quantitative comparison of the segmentation results of each organ in the TD dataset, presented as mean with 95% confidence intervals (color overlay).

Organ	MCC (%) [†]				HD ₉₅ (mm) [‡]			
	nnU-Net	PointRend	PolySnake	Ours	nnU-Net	PointRend	PolySnake	Ours
Body	94.36	92.60	94.12	94.97	0.94	2.63	1.47	0.97
	[94.05, 94.67]	[92.04, 93.16]	[93.61, 94.62]	[94.77, 95.17]	[0.92, 0.96]	[2.22, 3.05]	[1.27, 1.66]	[0.94, 1.00]
Kidney	88.43	88.88	92.96	93.21	1.11	1.03	0.62	0.48
	[87.02, 89.85]	[88.62, 89.14]	[92.49, 93.43]	[92.73, 93.68]	[0.99, 1.23]	[1.01, 1.05]	[0.59, 0.66]	[0.46, 0.50]
Spleen	90.09	78.68	88.81	91.52	0.93	1.35	0.68	0.51
	[89.42, 90.77]	[76.61, 80.75]	[87.81, 89.81]	[90.97, 92.08]	[0.85, 1.02]	[1.22, 1.48]	[0.64, 0.72]	[0.48, 0.54]
Vessel	/	/	/	84.17	/	/	/	0.34
				[82.94, 85.41]				[0.31, 0.37]
Spine	92.45	88.83	92.68	94.29	0.36	0.55	0.32	0.29
	[92.07, 92.82]	[87.73, 89.93]	[91.87, 93.48]	[93.76, 94.82]	[0.35, 0.38]	[0.50, 0.61]	[0.30, 0.35]	[0.26, 0.32]
Average	91.33	87.25	92.30	93.44	0.85	1.29	0.94	0.51
	[90.77, 91.89]	[86.39, 88.10]	[92.03, 92.58]	[93.20, 93.68]	[0.79, 0.90]	[1.17, 1.40]	[0.82, 1.06]	[0.50, 0.53]

animal breathing. Next, we demonstrate continuous monitoring and quantitative analysis of endogenous biomarkers in specific organs. For instance, we calculate the sO_2 distribution of the left kidney at different time and display it with a color overlay (Fig. 5b). The result reveals that sO_2 level near the right-side artery increases over time. In Fig. 6, we perform a quantitative analysis of the organ area continuously observed within 15 s from the segmentation results of the SFE-GCN model in the TD dataset. It can be observed that mice body and soft organs (kidneys and spleen) exhibit regular relaxation with respiratory movements under anesthesia, while the solid spine and small vessel areas remain relatively stable. This consistency with actual animal organ physiology highlights the accuracy and utility of our method.

4.2. Results on space domain continuous PAT images

Segmentation performance. The qualitative comparison of the segmentation results on the SD dataset is shown in Fig. 7. Compared with the visual outputs from the other methods, segmentation contour predicted by SFE-GCN not only aligns closely with the ground truth but also exhibit superior anatomical plausibility. Mask-based methods (UNet and PointRend) are affected by disconnected structures and mask holes (white arrows), especially in the liver which is difficult to segment. As a contour-based method, PolySnake obtains smoothly closed segmentation contour, yet it suffers from classification error for morphologically similar organs, e.g., misclassifying the vessel as a spine (red arrow). In contrast, our method benefits from the shared

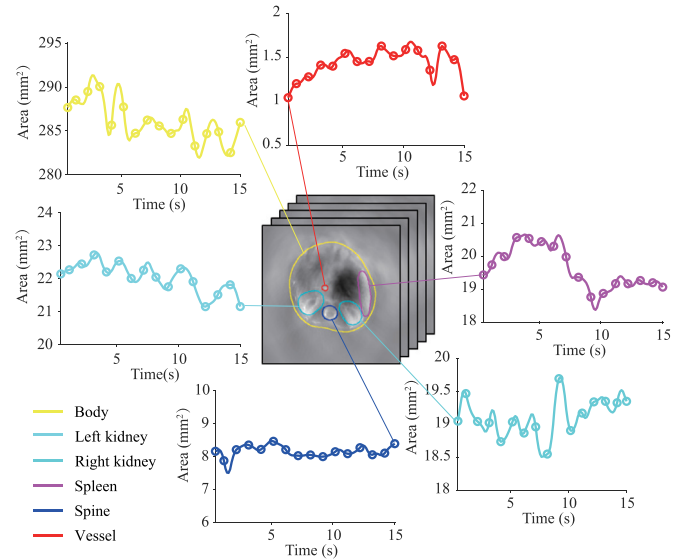


Fig. 6. Quantitative analysis of organ area continuously monitored within 15 s in TD dataset.

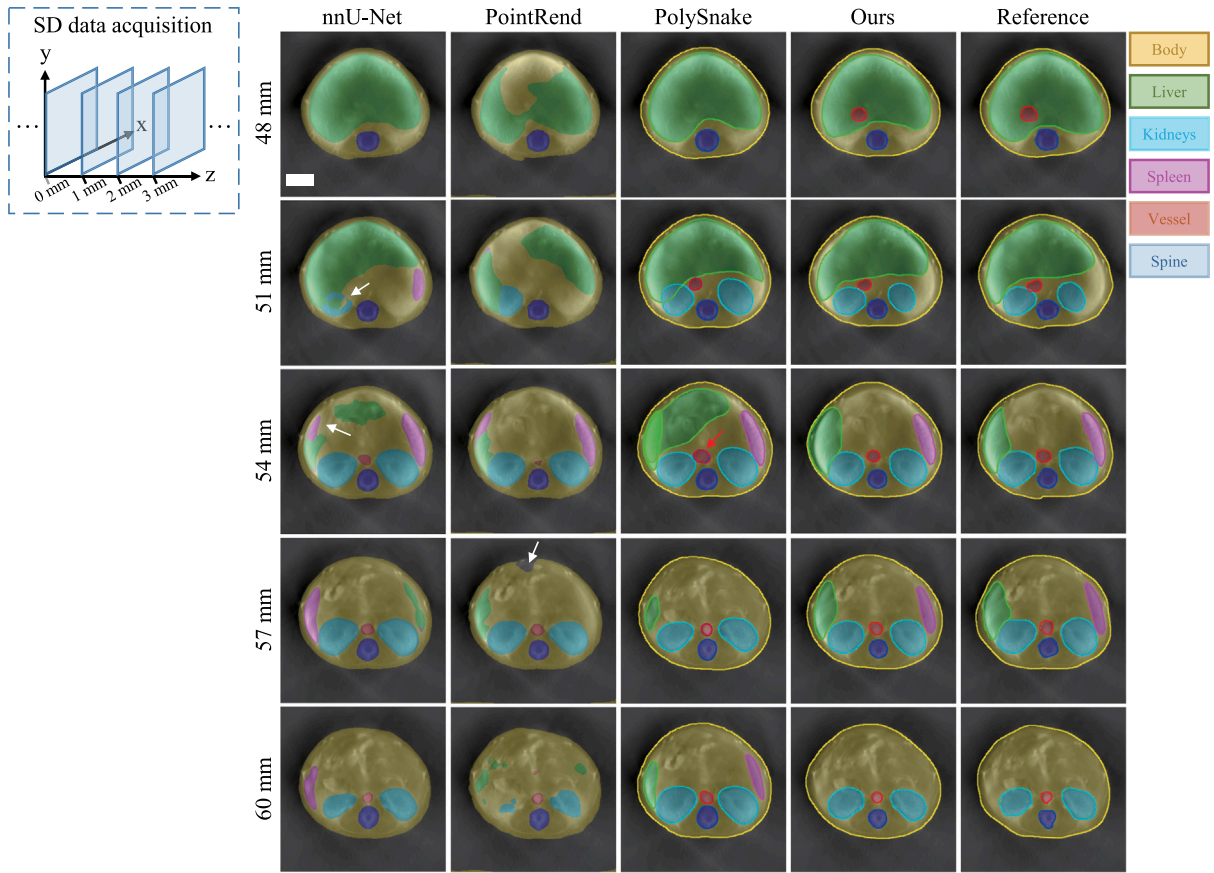


Fig. 7. Segmentation results of the space-continuous imaging experiment (SD dataset). Segmentations produced by mask-based methods are shown with filled semi-transparent regions. Segmentations of contour-based methods and human expert annotations are indicated with opaque lines. Scale bar: 5 mm.

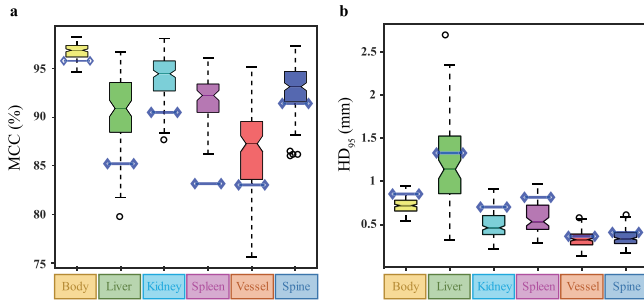


Fig. 8. Quantitative results of each organ in the SD dataset. a, b: Box plots of MCC and HD_{95} scores per organ; each box extends from lower to upper quartile values of data, with a black line at the median; the whiskers extend to the outmost data point within 1.5× the interquartile range; the outliers outside the whiskers are shown as red circles; the blue lines represent the baseline trained with a single slice.

information of the continuous image sequence and achieves the most reasonable predicted contour while maintaining high classification and segmentation accuracy. The quantitative results on the SD dataset is shown in Table S1 in the supplemental material.

To quantitatively assess the performance of SFE-GCN on individual organs, we computed the segmentation accuracy for six organs in the SD dataset, as shown in Fig. 8. For the SD dataset, the median MCC scores produced by SFE-GCN are above the 90% for the body, liver, kidneys, spleen, and spine (Fig. 8a). Interquartile ranges from 85% to 95%, suggesting that the organs are consistently segmented with high

precision. Except for the challenging liver boundary, HD_{95} scores of the other organs are lower than 1 mm (Fig. 8b), indicating that the predicted contours closely match the human annotations. To evaluate the performance of the context structural feature fusion module in improving segmentation accuracy, we trained the SFE-GCN model in a single-slice input mode as the baseline. The median quantitative scores of its segmentation results are represented by a blue line (Fig. 8a and b). The median MCCs of baseline are consistently inferior to those of our method, while the median HD_{95} scores are higher than the upper quartile of SFE-GCN. It highlights the efficacy of Contextual structure feature fusion in enhancing organ structure information and improving segmentation accuracy.

LF correction experiment. The results of the LF correction experiment using SD dataset are shown in Fig. 9. In the original PAT image, the light attenuation effect can be clearly seen, i.e., the PAT signal intensity attenuated with increased imaging depth. Previous segmentation-based LF correction methods only provides the overall segmentation of the animal body and a uniform absorption coefficient distribution μ_a^{body} . This body-only segmentation method moderately ameliorates the light attenuation and enhances the visual effect of the PAT image. The proposed multi-organ segmentation method enables organ-wise delineation of absorption distribution μ_a^{organs} and fosters a more realistic LF map. The image exhibits uniform intensity after organ-wise LF correction, providing enhanced clarity of internal structures and abdominal organs. Furthermore, deep central features such as the A/IVC exhibit improved signal intensity, and the liver and kidneys present more uniformly distributed signal intensities.

To further evaluate the image quality before and after LF correction, we plotted the intensity profiles (Fig. 10a) of the resulting PAT

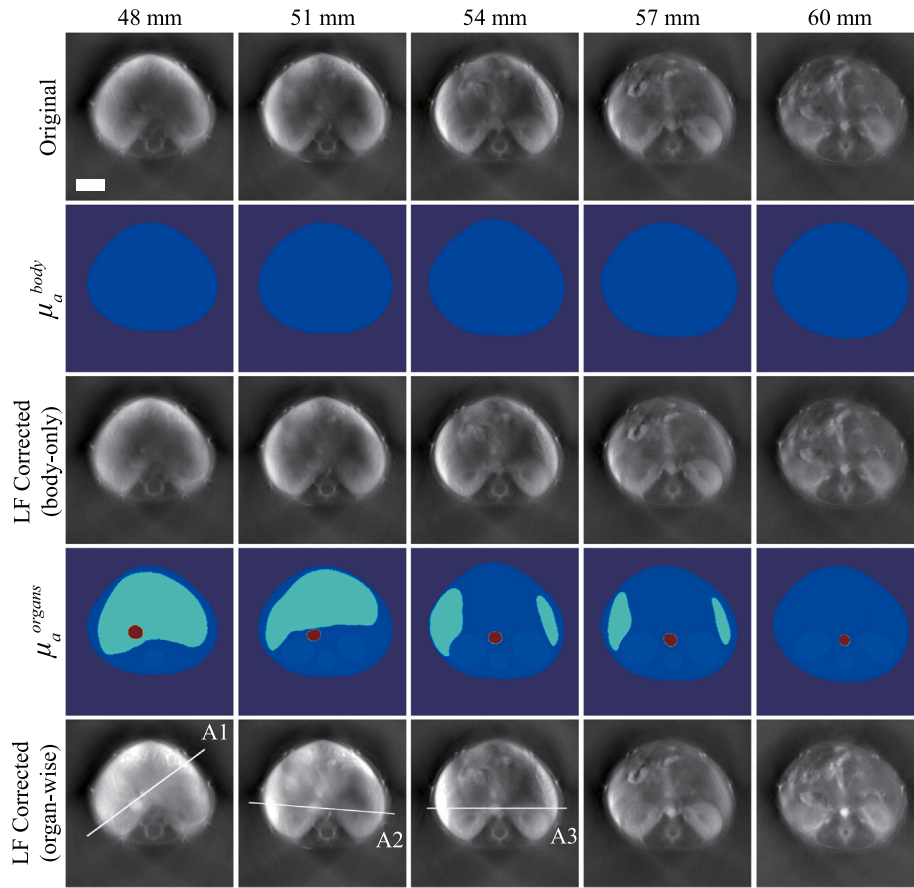


Fig. 9. Qualitative comparison results of LF correction experiments on the SD dataset. μ_a^{body} represents absorption coefficient map obtained by body-only segmentation. μ_a^{organs} represents the organ-level absorption coefficient map obtained by SFE-GCN segmentation. Scale bar: 5 mm.

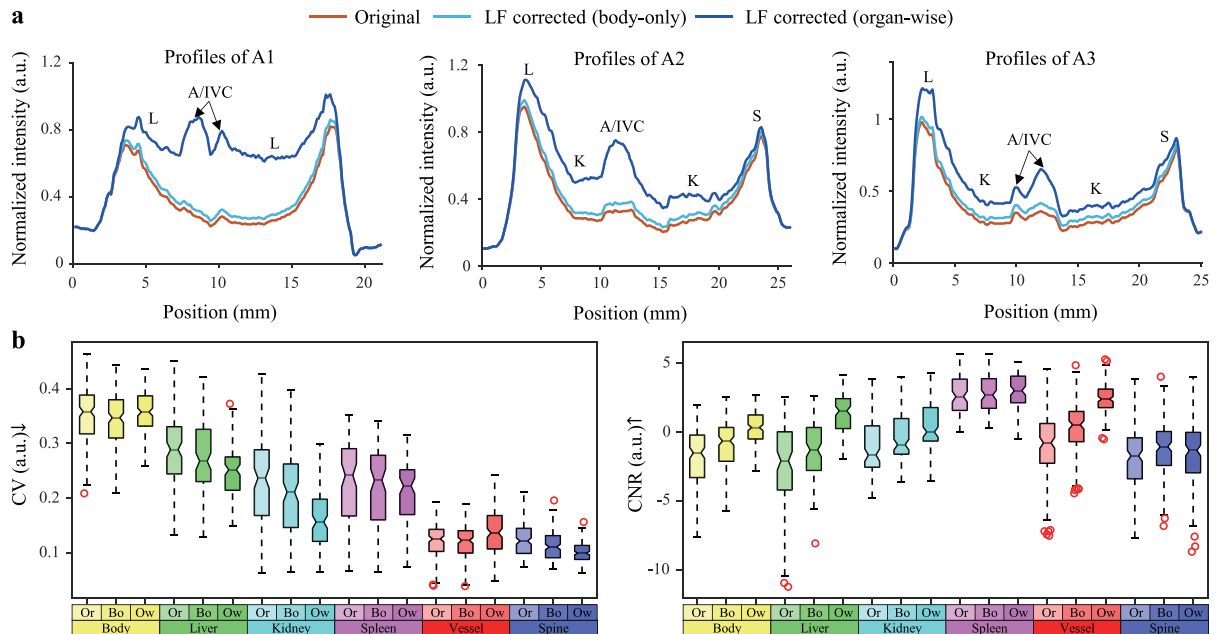
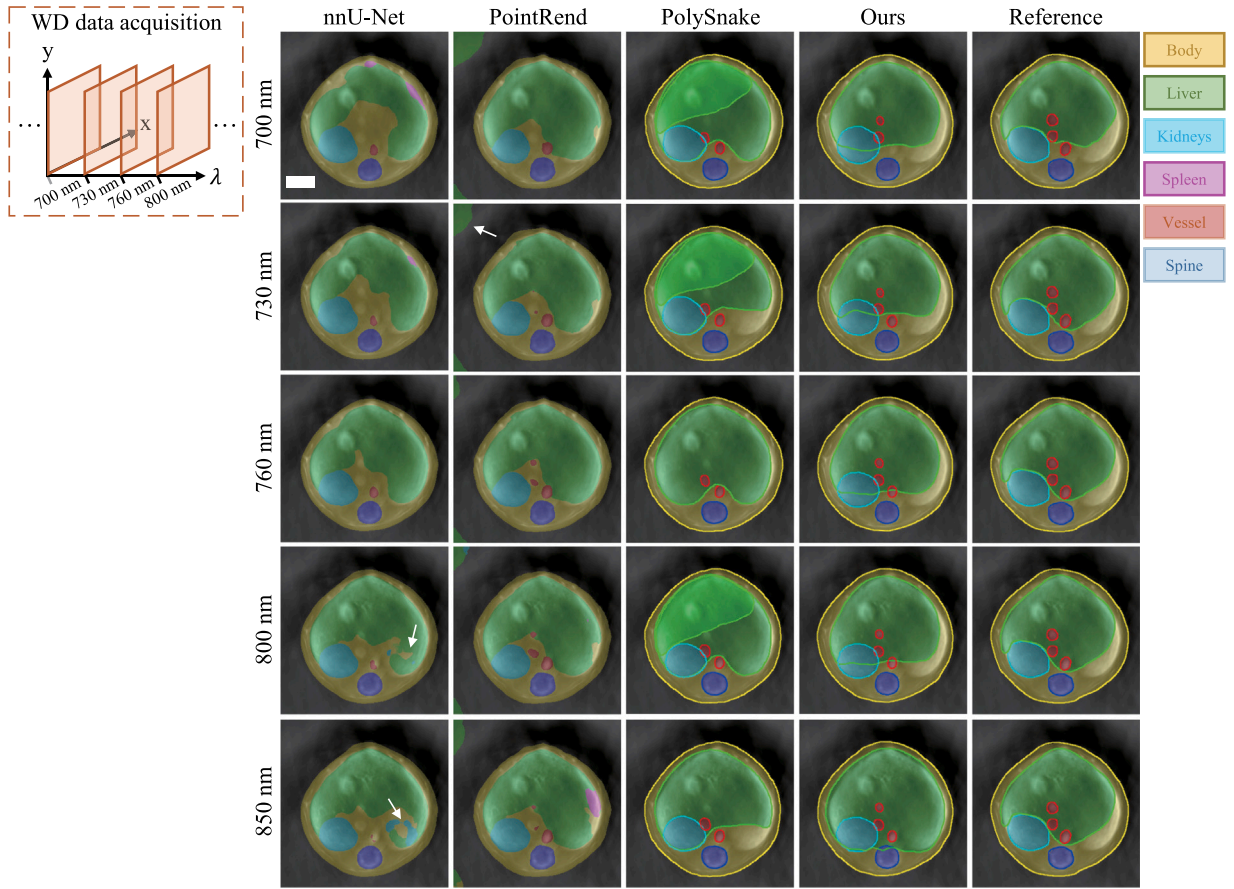


Fig. 10. a: The intensity profiles of the original PAT image, the body-only LF corrected image and organ-wise LF corrected image plotted along the white lines A1, A2, and A3 in Fig. 9. b: CV and CNR scores for each organ of PAT images before and after LF correction using SD datasets. Each box extends from lower to upper quartile values of data, with a black line at the median; the whiskers extend to the outmost data point within 1.5× the interquartile range. The outliers outside the whiskers are shown as red circles. Description in a: L: liver, A/IVC: artery/inferior vena cava; K: kidney; S: spleen. Abbreviations in b: Or: original PAT images; Bo: body-only LF corrected images; Ow: organ-wise LF corrected images.

Table 4

Quantitative comparison of the segmentation results of each organ in the WD dataset, presented as mean with 95% confidence intervals (color overlay).

Organ	MCC (%) [†]				HD ₉₅ (mm) [‡]			
	nnU-Net	PointRender	PolySnake	Ours	nnU-Net	PointRender	PolySnake	Ours
Body	95.56 [95.33, 95.80]	95.66 [95.46, 95.86]	95.89 [95.36, 96.41]	95.34 [95.20, 95.49]	0.87 [0.84, 0.91]	0.88 [0.84, 0.92]	0.59 [0.57, 0.60]	0.90 [0.88, 0.92]
Liver	71.53 [68.21, 74.86]	76.05 [73.64, 78.47]	67.70 [63.63, 71.77]	76.11 [72.83, 79.39]	3.74 [3.34, 4.14]	5.22 [4.87, 5.57]	3.74 [3.30, 4.17]	1.67 [1.43, 1.92]
Kidney	84.16 [81.88, 86.45]	82.63 [80.06, 85.20]	86.06 [83.66, 88.46]	88.68 [87.18, 90.18]	1.03 [0.95, 1.10]	1.12 [1.02, 1.23]	5.24 [5.08, 5.40]	0.70 [0.67, 0.72]
Spleen	77.44 [74.51, 80.37]	55.29 [50.34, 60.23]	76.70 [73.00, 80.40]	87.05 [86.39, 87.71]	1.37 [1.24, 1.50]	1.22 [1.13, 1.32]	0.91 [0.83, 1.00]	0.90 [0.84, 0.97]
Vessel	53.09 [48.57, 57.61]	59.15 [56.02, 62.27]	54.20 [50.51, 57.89]	79.35 [76.94, 81.76]	0.33 [0.28, 0.37]	0.48 [0.44, 0.53]	0.92 [0.82, 1.01]	0.26 [0.24, 0.28]
Spine	91.92 [91.64, 92.20]	87.23 [86.28, 88.19]	88.71 [87.51, 89.91]	91.73 [91.43, 92.04]	0.45 [0.44, 0.46]	0.60 [0.58, 0.62]	0.45 [0.44, 0.47]	0.43 [0.42, 0.44]
Average	82.13 [81.10, 83.16]	80.09 [79.03, 81.16]	81.18 [79.95, 82.41]	88.42 [87.85, 88.99]	1.64 [1.54, 1.75]	2.04 [1.91, 2.16]	2.26 [2.19, 2.33]	0.80 [0.77, 0.82]

**Fig. 11.** Segmentation results of the spectrum-continuous imaging experiment (WD dataset). Segmentations produced by mask-based methods are shown with filled semi-transparent regions. Segmentations of contour-based methods and human expert annotations are indicated with opaque lines. Scale bar: 5 mm.

images along the positions of the three white lines (A1, A2, and A3) in the Fig. 9. In the A1 and A3 profiles of the original PAT image, the signal in the superficial layers of the liver and kidney is higher than that in the deeper layers. This non-uniform intensity distribution also exists in the body-only corrected image. Conversely, our organ-wise LF correction method enhances deep signals within these organs and improves signal uniformity. In PAT images, blood-rich arteries or inferior vena cava have high optical absorption and should therefore show stronger signals. However, both the original PAT image and the body-only corrected image fail to accurately depict this phenomenon. In our organ-wise corrected image, the signal amplitude of A/IVC is significantly enhanced in all A1, A2 and A3 profiles. Importantly, the

signal intensity of central arteries is corrected to the same level in both A1 and A2 profiles, indicating close alignment between the optical coefficient distribution obtained by the organ-wise correction method and that of real animal organs. To evaluate the enhancement of the visual quality after correction, we calculated visual quality evaluation scores (CV and CNR) for each organ on the images in SD datasets (Fig. 10b). Compared to the body-only corrected image, organ-wise corrected image exhibits decreased median CV scores on the liver (0.28 vs. 0.25) and kidneys (0.21 vs. 0.17), indicating an enhancement in intra-organ signal consistency. Furthermore, the CNR score of our method significantly improved across all organs, indicating enhanced image contrast after organ-wise correction.

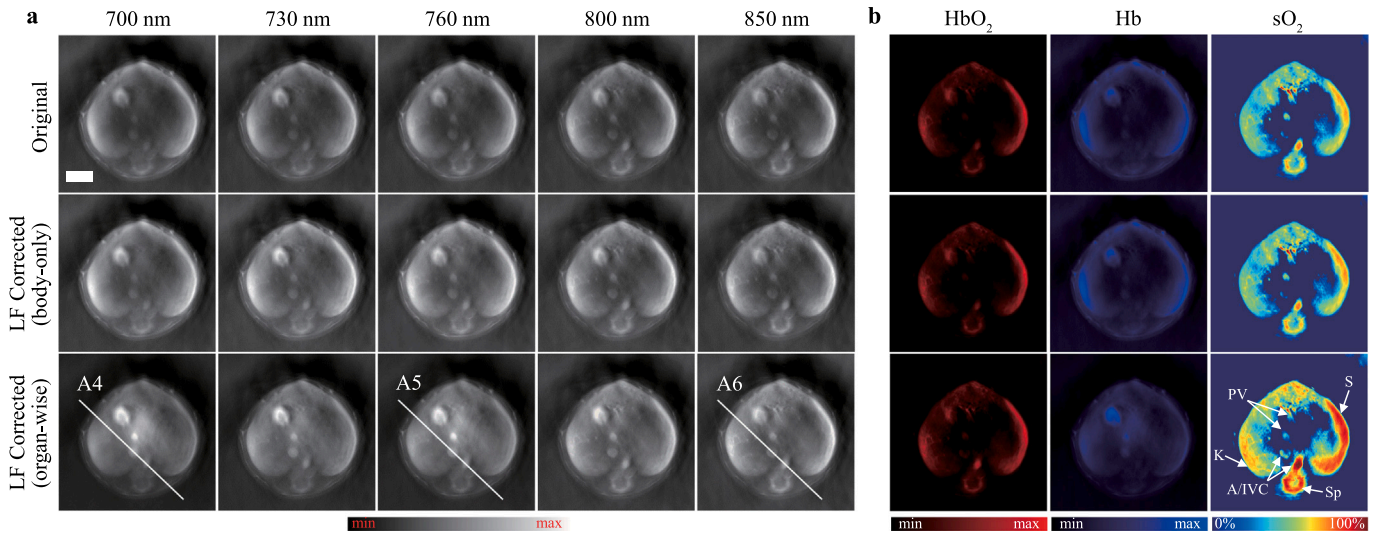


Fig. 12. a: Qualitative comparison results of LF correction experiments on the WD dataset. b: Spectral un-mixing results. HbO₂: oxygenated hemoglobin, Hb: deoxygenated hemoglobin, and sO₂: blood oxygen saturation. A: artery; IVC: inferior vena cava; PV: portal vein; K: kidney; S: spleen; Sp: spine; Scale bar: 5 mm.

4.3. Results on spectrum domain continuous PAT images

Segmentation performance. The qualitative comparison of the segmentation results of the WD dataset is shown in Fig. 11. As can be seen, SFE-GCN provides predictions closest to the ground truth across PAT images at five distinct wavelengths. Similar to SD dataset, UNet and PointRend exhibit predictions with disconnected structures and small holes (white arrows). PolySnake repetitively segments the liver (at 700, 730 and 800 nm). Although our method also deviates from the reference annotation in some places, it consistently achieves segmentation aligning with the anatomical characteristics of organs, especially at the three vessels. Noting that the penetration ability of laser attenuates as the spectral wavelength decreases, and thus substantial loss of structural information in PAT images is captured at shorter wavelengths. However, our SFE-GCN fuses structural enhancement information from multiple wavelengths, and thus still enables robust segmentation even for PAT images at 700 nm and 730 nm.

For the WD dataset, we quantitatively compare the performance scores of the four methods on all organs in Table 4. For the clearest body and spine, PolySnake and nnU-Net yield the highest MCC scores (95.89% and 91.92%) respectively. For the vessels and kidneys with small sizes and weakly boundaries, our method achieves the highest performance scores. The comparison results of the HD₉₅ scores demonstrates that SFE-GCN can robustly obtain segmentation contours closest to the anatomical boundaries of organs.

LF correction experiment. The results of the LF correction experiment on WD dataset imaged at five different wavelengths are shown in Fig. 12a. Comparative analysis with the original PAT image and the body-only corrected image reveals significant enhancements in both signal intensity uniformity and organ structure clarity in the organ-wise corrected image. It is worth noting that effective restoration of deep tissue information is achieved in the 700 nm PAT image, which has the weakest laser penetration. To further evaluate the image quality before and after LF correction, we plot the intensity profiles along three white lines (A4, A5, and A6) in the image at 700, 760 and 850 nm. The profiles demonstrate that organ-wise LF correction enhances the deep signals of the liver and two A/IVC (Fig. 13a). Importantly, it significantly improves signal uniformity within the liver, while correcting the signal amplitudes of the two A/IVCs to a consistent level. The visual quality evaluation scores of each organ on WD dataset before and

after correction are shown in (Fig. 13b). It can be seen that the organ-wise corrected image achieves substantial image quality enhancement compared with the body-only corrected image. Notably, in liver with the lowest structural resolution, our approach achieves significant improvement in CV and CNR scores, highlighting the significant potential of the organ-wise LF correction method to enhance PAT image quality.

Spectral un-mixing experiments. Fig. 12b shows the HbO₂, Hb and sO₂ distribution in the abdominal region. During in vivo imaging, the Hb concentration of hypoxic portal vein (PV) should be higher than in other tissues, and the HbO₂ should be highest in the artery (A). However, due to laser energy attenuation, these features are not correctly reflected in the original PAT image. It further causes errors in the calculation of sO₂ value. In Fig. 12b, the body-only correction method fails to significantly enhance deep signals, resulting in corrected HbO₂, Hb and sO₂ distributions that closely resemble the original image. In contrast, the organ-wise correction method enhances HbO₂ and Hb concentrations in the imaging depth by rendering all blood vessels clearly visible. The sO₂ signal is also enhanced in oxygen-rich organs such as spleen. In conclusion, the spectral un-mixing results obtained through the organ-wise correction method accurately visualize and enhance information regarding deep endogenous markers such as A/IVC, PV, and spleen.

4.4. Hyper-parameter selection results

The hyper-parameter L is critical in determining the number of adjacent slices considered in the input image sequence during training. We establish a baseline with $L = 0$ and conduct comparative experiments with $L = 1, 2, 3, 4$, respectively. We performed comparative experiments using the SD dataset, and the quantitative results are shown in Table 5. The results show that $L = 2$ has the best segmentation performance with the highest MCC. When $L > 2$, the images from distant slices exhibit low structural similarity, potentially misleading SFE-GCN and leading to diminished segmentation performance. Consequently, in all other experiments we chose $L = 2$.

In addition, we conduct additional experiments to investigate whether the number of vertices K representing the contour affects model performance. In theory, larger values of K can capture finer boundary details (lower HD₉₅), while smaller values demand lower computational cost. Table 6 shows that the performance of the model is similar when the number of vertices is selected differently, and the larger

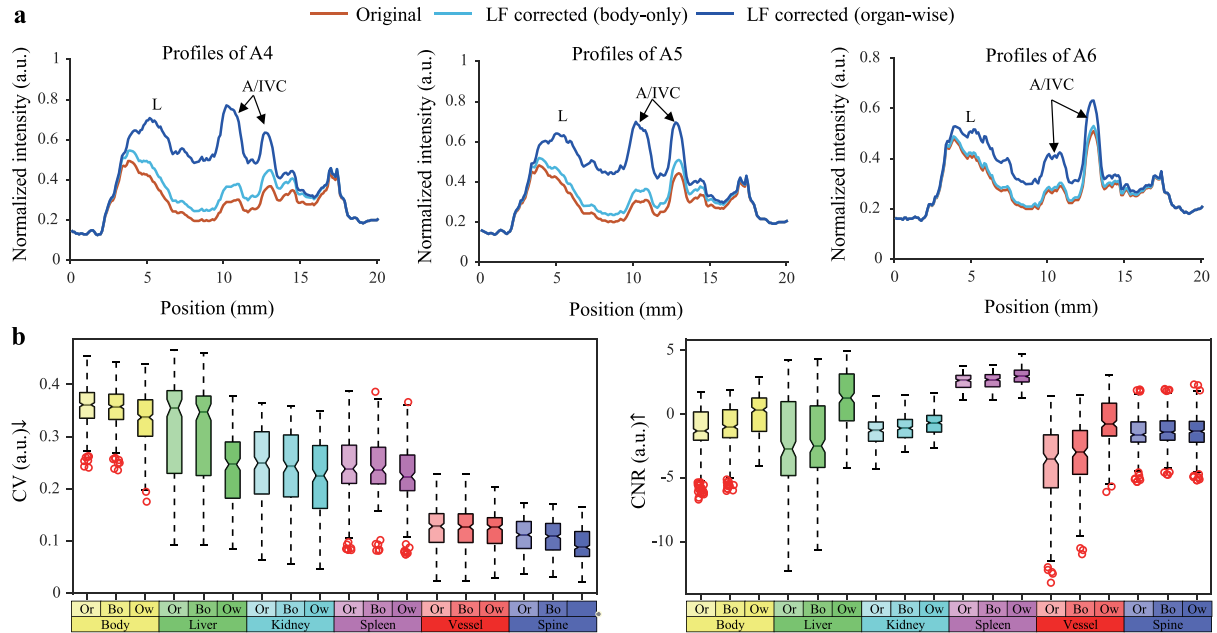


Fig. 13. a: The intensity profiles of the original PAT image, the body-only LF corrected image and organ-wise LF corrected image plotted along the white lines A4, A5 and A6 in Fig. 12. b: CV and CNR scores for each organ of PAT images before and after LF correction using WD datasets. Each box extends from lower to upper quartile values of data, with a black line at the median; the whiskers extend to the outmost data point within 1.5x the interquartile range. The outliers outside the whiskers are shown as red circles. Description in a: L: liver, A/IVC: artery/inferior vena cava. Abbreviations in b: Or: original PAT images; Bo: body-only LF corrected images; Ow: organ-wise LF corrected images.

Table 5

Model performance with different size (L) of the adjacent image sequence.

Value of L	MCC (%)	HD ₉₅ (mm)
0	90.73 [90.36, 91.11]	0.73 [0.68, 0.78]
1	91.40 [91.05, 91.74]	0.74 [0.68, 0.80]
2	92.21 [91.94, 92.48]	0.64 [0.60, 0.69]
3	91.69 [91.41, 91.98]	0.69 [0.65, 0.73]
4	91.68 [91.32, 92.04]	0.70 [0.64, 0.75]

Table 6

Model performance with different number of vertices (K).

No. of K	MCC (%)	HD ₉₅ (mm)
16	91.60 [91.32, 91.88]	1.11 [1.06, 1.15]
32	91.61 [91.34, 91.87]	0.78 [0.74, 0.82]
64	92.21 [91.94, 92.48]	0.64 [0.60, 0.69]
128	92.60 [92.25, 92.94]	0.62 [0.56, 0.67]
256	92.17 [91.92, 92.42]	0.60 [0.56, 0.64]

the number of nodes, the higher the performance. As the performance improvement gains become marginal beyond the $K = 64$, we choose $K = 64$ in this work.

4.5. Generalization performance

SFE-GCN is not constrained to a specific imaging scene and specific imaging position. As shown in Table 7, we trained the model on TD, SD and WD datasets and subsequently tested it on the three respective imaging domain datasets. The results show that although there is a certain degree of decline in MCC scores for cross-domain testing set, the model's generalization performance on cross-domain PAT images remains satisfactory. In addition, the model can be easily retrained from scratch for organ segmentation in other positions, and our pre-trained model speeds up this process even further.

Table 7

Evaluation of cross-scenario generalization ability of SFE-GCN.

Training \ Testing	TD dataset	SD dataset	WD dataset
TD dataset	93.44 [93.20, 93.68]	87.27 [86.16, 88.38]	91.95 [91.54, 92.37]
SD dataset	90.28 [89.64, 90.93]	92.21 [91.94, 92.48]	88.44 [87.50, 89.39]
WD dataset	86.77 [86.02, 87.52]	83.82 [82.87, 84.76]	88.42 [87.85, 88.99]

5. Discussion

Multi-dimensional continuous PAT images encapsulate rich information of biological tissue across temporal, spatial, and spectral domains. Harnessing the correlation within these domains holds the potential for more accurate multi-parametric physiological analysis. The proposed SFE-GCN, as an end-to-end instance segmentation pipeline, is designed to facilitate multi-organ segmentation in abdominal PAT images by fusing multi-dimensional continuous image information. As demonstrated in this work, our SFE-GCN enables the extraction of quantitative parameters from continuous images, such as the size of organ, chromophore localization, and the distribution of endogenous biomarkers. The accurate organ delineation provided by SFE-GCN will be helpful in many field of PAT research, including organ damage study (Guevara et al., 2013), tumor research (Zhang et al., 2019a), image quality enhancement (Liang et al., 2022; Mandal et al., 2016), visualization improvement of endogenous or exogenous absorbers (Zhang et al., 2022a) and hemodynamic analysis under oxygen challenge (Chen et al., 2022). Our organ-based multi-parametric image analysis experiments elucidate more accurate bio-optical information by using temporal-spatial-spectral relationships in continuous PAT image sequences.

The structural information inherent in the time, space, and spectral domains of continuous PAT images allows for more accurate image classification and segmentation. As evaluated in an extensive abdominal PAT validation experiment, our SFE-GCN shows effective learning capability of the shared structural information of adjacent images with

segmentation quality matching or exceeding state-of-the-art methods. If the boundary is invisible or poorly defined, such as the liver, our SFE-GCN can leverage the knowledge of the boundary shape to find highly feasible separations. Furthermore, SFE-GCN has high tolerance for overlapping organs (e.g., liver and left kidney in Fig. 11), which avoids inconsistencies that may occur in mask-based methods. The continuous and closed nature of contours captured by our ResGCN module contributes to this tolerance. In fact, the superior performance of SFE-GCN is rooted in a design resembling human visual annotation, i.e., adjacent frame observation and continuous contour annotation. It means that further exploration of human conscious activities (such as attention) during labeling will likely continue to improve the segmentation performance.

6. Conclusion

In this paper, we propose an end-to-end multi-organ instance segmentation method named SFE-GCN for abdominal PAT images. By incorporating a multi-scale feature extractor and a GCN-based contour detector, the method fuses structural enhancement information from continuous PAT image sequence, whether captured in time, space, or spectrum domains. On three distinct continuous imaging scenarios, SFE-GCN matches or exceeds the performance of state-of-the-art segmentation models. With this method, we further demonstrate dynamic analysis of organ size changes, organ-wise light fluence correction, and segmentation-enhanced spectral un-mixing. The results show that organ-level segmentation provided by SFE-GCN may significantly aid various preclinical PAT imaging tasks by facilitating quantitative multi-parametric analysis.

CRediT authorship contribution statement

Zhichao Liang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shuangyang Zhang:** Resources, Data curation. **Zongxin Mo:** Formal analysis, Data curation. **Xiaoming Zhang:** Formal analysis, Data curation. **Anqi Wei:** Formal analysis. **Wufan Chen:** Resources, Funding acquisition. **Li Qi:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.media.2024.103402>.

Data availability

Our research code has been uploaded to Github, the access link is attached in the manuscript.

References

- Alexandrakis, G., Rannou, F.R., Chatzioannou, A.F., 2005. Tomographic bioluminescence imaging by use of a combined optical-PET (OPET) system: A computer simulation feasibility study. *Phys. Med. Biol.* 50 (17), 4225. <http://dx.doi.org/10.1088/0031-9155/50/17/021>.
- Chen, Z., Gezginer, I., Augath, M.A., Ren, W., Liu, Y.H., Ni, R., Deán-Ben, X.L., Razansky, D., 2022. Hybrid magnetic resonance and optoacoustic tomography (MROT) for preclinical neuroimaging. *Light Sci. Appl.* 11 (1), 332. <http://dx.doi.org/10.1038/s41377-022-01026-w>.
- Cox, B.T., Arridge, S.R., Köstli, K.P., Beard, P.C., 2006. Two-dimensional quantitative photoacoustic image reconstruction of absorption distributions in scattering media by use of a simple iterative method. *Appl. Opt.* 45 (8), 1364–1366. <http://dx.doi.org/10.1364/AO.45.001866>.
- Cox, B., Laufer, J.G., Arridge, S.R., Beard, P.C., 2012. Quantitative spectroscopic photoacoustic imaging: a review. *J. Biomed. Opt.* 17 (6), 061202. <http://dx.doi.org/10.1117/1.jbo.17.6.061202>.
- Deán-Ben, X.L., Ford, S.J., Razansky, D., 2015. High-frame rate four dimensional optoacoustic tomography enables visualization of cardiovascular dynamics and mouse heart perfusion. *Sci. Rep.* 5 (November 2014), 1–9. <http://dx.doi.org/10.1038/srep10133>.
- Feng, H., Zhou, W., Yin, Y., Deng, J., Sun, Q., Li, H., 2023. Recurrent contour-based instance segmentation with progressive learning. pp. 1–11. [arXiv:2301.08898](https://arxiv.org/abs/2301.08898).
- Guevara, E., Berti, R., Londono, N., Xie, N., Bellec, P., Lesage, R., Lodygensky, G.A., 2013. Imaging of an inflammatory injury in the newborn rat brain with Photoacoustic tomography. *PLoS ONE* 8 (12), e83045. <http://dx.doi.org/10.1371/journal.pone.0083045>.
- Hao, D., Li, H., 2023. A graph-based edge attention gate medical image segmentation method. *IET Image Process.* 17 (7), 2142–2157. <http://dx.doi.org/10.1049/ipr2.12780>.
- Huang, R., Lin, M., Dou, H., Lin, Z., Ying, Q., Jia, X., Xu, W., Mei, Z., Yang, X., Dong, Y., Zhou, J., Ni, D., 2022. Boundary-rendering network for breast lesion segmentation in ultrasound images. *Med. Image Anal.* 80, 102478. <http://dx.doi.org/10.1016/j.media.2022.102478>.
- Isensee, F., Jaeger, P.F., Kohl, S.A., Petersen, J., Maier-Hein, K.H., 2021. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nat. Methods* 18 (2), 203–211. <http://dx.doi.org/10.1038/s41592-020-01008-z>.
- Ivankovic, I., Mercep, E., Schmedt, C.G., Deán-Ben, X.L., Razansky, D., 2019. Real-time volumetric assessment of the human carotid artery: Handheld multispectral optoacoustic tomography. *Radiology* 291 (1), 45–50. <http://dx.doi.org/10.1148/radiol.2019181325>.
- Keshava, N., Mustard, J.F., 2002. Spectral unmixing. *IEEE Signal Process. Mag.* 19 (1), 44–57. <http://dx.doi.org/10.1109/79.974727>.
- Kirillov, A., Wu, Y., He, K., Girshick, R., 2020. Pointrend: Image segmentation as rendering. In: *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. pp. 9799–9808. <http://dx.doi.org/10.1109/CVPR42600.2020.00982>.
- Knieling, F., Neufert, C., Hartmann, A., Claussen, J., Ulrich, A., Egger, C., Vetter, M., Fischer, S., Pfeifer, L., Hagel, A., Kielisch, C., Görtz, R.S., Wildner, D., Engel, M., Röther, J., Uter, W., Siebler, J., Atreya, R., Rascher, W., Strobel, D., Neurath, M.F., Waldner, M.J., 2017. Multispectral optoacoustic tomography for assessment of Crohn's disease activity. *N. Engl. J. Med.* 376 (13), 1292–1294. <http://dx.doi.org/10.1056/nejmc1612455>.
- Law, H., Deng, J., 2020. CornerNet: Detecting objects as paired keypoints. *Int. J. Comput. Vis.* 128, 642–656. <http://dx.doi.org/10.1007/s11263-019-01204-1>.
- Li, X., Ge, J., Zhang, S., Wu, J., Qi, L., Chen, W., 2022. Multispectral interlaced sparse sampling photoacoustic tomography based on directional total variation. *Comput. Methods Programs Biomed.* 214, 106562. <http://dx.doi.org/10.1016/j.cmpb.2021.106562>.
- Li, X., Qi, L., Zhang, S., Huang, S., Wu, J., Lu, L., Feng, Y., Feng, Q., Chen, W., 2019. Model-based optoacoustic tomography image reconstruction with non-local and sparsity regularizations. *IEEE Access* 7, 102136–102148. <http://dx.doi.org/10.1109/access.2019.2930650>.
- Li, M., Tang, Y., Yao, J., 2018. Photoacoustic tomography of blood oxygenation: A mini review. *Photoacoustics* 10, 65–73. <http://dx.doi.org/10.1016/j.pacs.2018.05.001>.
- Liang, Z., Zhang, S., Wei, A., Liu, Z., Wang, Y., Hu, H., Chen, W., Qi, L., 2024. CylindGCN: Cylindrical structures segmentation in 3D biomedical optical imaging by a contour-based graph convolutional network. *Comput. Med. Imaging Graph.* 111 (June 2023), 102316. <http://dx.doi.org/10.1016/j.compmedimag.2023.102316>.
- Liang, Z., Zhang, S., Wu, J., Li, X., Zhuang, Z., Feng, Q., Chen, W., Qi, L., 2022. Automatic 3-D segmentation and volumetric light fluence correction for photoacoustic tomography based on optimal 3-D graph search. *Med. Image Anal.* 75, 102275. <http://dx.doi.org/10.1016/j.media.2021.102275>.
- Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P., 2020. Focal loss for dense object detection. *IEEE Trans. Pattern Anal. Mach. Intell.* 42 (2), 2980–2988. <http://dx.doi.org/10.1109/TPAMI.2018.2858826>.
- Lv, J., Xu, Y., Xu, L., Nie, L., 2021. Quantitative functional evaluation of liver fibrosis in mice with dynamic contrast-enhanced photoacoustic imaging. *Radiology* 300 (1), 89–97. <http://dx.doi.org/10.1148/radiol.2021204134>.

- Ly, C.D., Nguyen, V.T., Vo, T.H., Mondal, S., Park, S., Choi, J., Vu, T.T.H., Kim, C.S., Oh, J., 2022. Full-view in vivo skin and blood vessels profile segmentation in photoacoustic imaging based on deep learning. *Photoacoustics* 25, 100310. <http://dx.doi.org/10.1016/j.pacs.2021.100310>.
- Mandal, S., Dean-Ben, X.L., Razansky, D., 2016. Visual quality enhancement in optoacoustic tomography using active contour segmentation priors. *IEEE Trans. Med. Imaging* 2209–2217. <http://dx.doi.org/10.1109/TMI.2016.2553156>.
- Manivasagan, P., Bharathiraja, S., Bui, N.Q., Jang, B., Oh, Y.O., Lim, I.G., Oh, J., 2016. Doxorubicin-loaded fucoidan capped gold nanoparticles for drug delivery and photoacoustic imaging. *Int. J. Biol. Macromol.* 91, 578–588. <http://dx.doi.org/10.1016/j.ijbiomac.2016.06.007>.
- Marin, D., He, Z., Vajda, P., Chatterjee, P., Tsai, S., Yang, F., Boykov, Y., 2019. Efficient segmentation: Learning downsampling near semantic boundaries. In: *Proceedings of the IEEE International Conference on Computer Vision*, Vol. 2019-Octob. pp. 2131–2141. <http://dx.doi.org/10.1109/ICCV.2019.00222>.
- Meng, Y., Zhang, H., Gao, D., Zhao, Y., Yang, X., Qian, X., Huang, X., Zheng, Y., 2021. BI-GCN: Boundary-aware input-dependent graph convolution network for biomedical image segmentation. *arXiv:2110.14775*.
- Moustakidis, S., Omar, M., Aguirre, J., Mohajerani, P., Ntzachristos, V., 2019. Fully automated identification of skin morphology in raster-scan optoacoustic mesoscopy using artificial intelligence. *Med. Phys.* 46 (9), 4046–4056. <http://dx.doi.org/10.1002/mp.13725>.
- Nikolay, S., 2024. mask2poly (Version 1.5.0.0) [software]. URL <https://www.mathworks.com/matlabcentral/fileexchange/32112-mask2poly>, MATLAB Central File Exchange.
- Peng, S., Jiang, W., Pi, H., Li, X., Bao, H., Zhou, X., 2020. Deep snake for real-time instance segmentation. In: *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*. pp. 8533–8542. <http://dx.doi.org/10.1109/CVPR42600.2020.00856>.
- Qi, L., Wu, J., Li, X., Zhang, S., Huang, S., Feng, Q., Chen, W., 2021. Photoacoustic tomography image restoration with measured spatially variant point spread functions. *IEEE Trans. Med. Imaging* 40 (9), 2318–2328. <http://dx.doi.org/10.1109/TMI.2021.3077022>.
- Razansky, D., Buehler, A., Ntzachristos, V., 2011. Volumetric real-time multispectral optoacoustic tomography of biomarkers. *Nat. Protoc.* 6, 1121–1129. <http://dx.doi.org/10.1038/nprot.2011.351>.
- Schellenberg, M., Dreher, K.K., Holzwarth, N., Isensee, F., Reinke, A., Schreck, N., Seitel, A., Tizabi, M.D., Maier-Hein, L., Gröhl, J., 2022. Semantic segmentation of multispectral photoacoustic images using deep learning. *Photoacoustics* 26, 100341. <http://dx.doi.org/10.1016/j.pacs.2022.100341>.
- Schweiger, M., Arridge, S., 2014. The Toast++ software suite for forward and inverse modeling in optical tomography. *J. Biomed. Opt.* 19 (4), 040801. <http://dx.doi.org/10.1117/1.jbo.19.4.040801>.
- Tang, K., Zhang, S., Liang, Z., Wang, Y., Ge, J., Chen, W., Qi, L., 2023. Advanced image post-processing methods for photoacoustic tomography: A review. *Photonics* 10 (7), <http://dx.doi.org/10.3390/photonics10070707>.
- Taruttis, A., Ntzachristos, V., 2015. Advances in real-time multispectral optoacoustic imaging and its applications. *Nat. Photonics* 9 (4), 219–227. <http://dx.doi.org/10.1038/nphoton.2015.29>.
- Tian, Z., Li, X., Zheng, Y., Chen, Z., Shi, Z., Liu, L., Fei, B., 2020. Graph-convolutional-network-based interactive prostate segmentation in MR images. *Med. Phys.* 47 (9), 4164–4176. <http://dx.doi.org/10.1002/mp.14327>.
- Yuan, A.Y., Gao, Y., Peng, L., Zhou, L., Liu, J., Zhu, S., Song, W., 2020. Hybrid deep learning network for vascular segmentation in photoacoustic imaging. *Biomed. Opt. Express* 11, 6445–6457. <http://dx.doi.org/10.1364/boe.409246>.
- Zhang, J., Chen, B., Zhou, M., Lan, H., Gao, F., 2019a. Photoacoustic image classification and segmentation of breast cancer: A feasibility study. *IEEE Access* 7, 5457–5466. <http://dx.doi.org/10.1109/ACCESS.2018.2888910>.
- Zhang, S., Liang, Z., Tang, K., Li, X., Zhang, X., Mo, Z., Wu, J., Huang, S., Liu, J., Zhuang, Z., Qi, L., Chen, W., 2023. In vivo co-registered hybrid-contrast imaging by successive photoacoustic tomography and magnetic resonance imaging. *Photoacoustics* 31, 100506. <http://dx.doi.org/10.1016/j.pacs.2023.100506>.
- Zhang, S., Liu, J., Liang, Z., Ge, J., Feng, Y., Chen, W., Qi, L., 2022a. Pixel-wise reconstruction of tissue absorption coefficients in photoacoustic tomography using a non-segmentation iterative method. *Photoacoustics* 28, 100390. <http://dx.doi.org/10.1016/j.pacs.2022.100390>.
- Zhang, S., Qi, L., Li, X., Liang, Z., Sun, X., Liu, J., Lu, L., Feng, Y., Chen, W., 2022b. MRI information-based correction and restoration of photoacoustic tomography. *IEEE Trans. Med. Imaging* 41 (9), 2543–2555. <http://dx.doi.org/10.1109/TMI.2022.3165839>.
- Zhang, S., Qi, L., Li, X., Liu, J., Huang, S., Wu, J., Liu, R., Feng, Y., Feng, Q., Chen, W., 2019b. Photoacoustic imaging of living mice enhanced with a low-cost contrast agent. *Biomed. Opt. Express* 10 (11), 5744–5754. <http://dx.doi.org/10.1364/boe.10.005744>.
- Zhou, X., Austin, U.T., Wang, D., Berkeley, U.C., Austin, U.T., 2019. Object as point. *arXiv:1904.07850*.
- Zhou, X., Koltun, V., Krähenbühl, P., 2020. Tracking Objects as Points. In: *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 12349 LNCS, pp. 474–490. http://dx.doi.org/10.1007/978-3-030-58548-8_28.